

EE 66
Fall 2026

Signals, Dynamics, and Information

Discussion 3B

1. DTFS Expansion

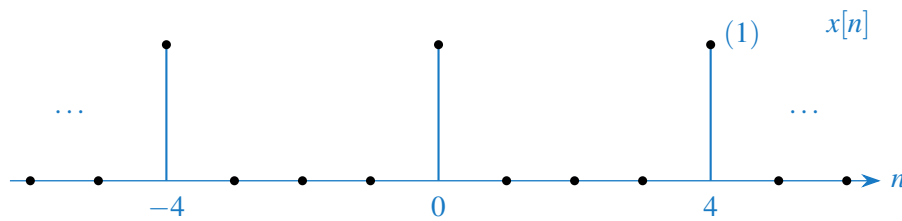
Determine the complex-exponential discrete-time Fourier series (DTFS) expansion for the signal $x : \mathbb{Z} \rightarrow \mathbb{R}$ described below, or explain why no such expansion exists. For the case where a DTFS expansion exists, be sure to identify the period p and the fundamental frequency ω_0 .

(a) In this problem, we introduce the Kronecker-Delta function δ . This function is defined as follows:

$$\delta[n] = \begin{cases} 1 & \text{if } n = 0 \\ 0 & \text{if } n \neq 0 \end{cases}$$

Let $x[n] = \sum_{\ell=-\infty}^{\infty} \delta[n-4\ell]$ be a signal, for all integers n . To start, provide a well-labeled plot of the signal x first, so you can also see its structure. Then, perform the DTFS expansion, if it exists.

Answer: This is a periodic impulse train.



The signal is periodic with fundamental period $p = 4$, which corresponds to a fundamental frequency of

$$\omega_0 = \frac{2\pi}{4} = \frac{\pi}{2}.$$

The DTFS coefficients are given by

$$X_k = \frac{1}{p} \sum_{n=0}^{p-1} x[n] e^{-ik\omega_0 n}, \quad k = 0, 1, 2, 3.$$

But only the $n = 0$ term is nonzero in the summation, so we have

$$X_k = \frac{1}{4} x[0] = \frac{1}{4} \quad k = 0, 1, 2, 3.$$

This result shows that all frequencies $0, \omega_0 = \pi/2, 2\omega_0 = \pi$, and $3\omega_0 = 3\pi/2$ contribute equally to the impulse train. That is,

$$x[n] = \frac{1}{4} \left(1 + e^{i\pi n/2} + e^{i\pi n} + e^{i3\pi n/2} \right).$$

(b) $x[n] = \cos\left(\frac{2\pi}{5}n\right), \quad \forall n.$

Answer: The signal is periodic with fundamental period $p = 5$, so its fundamental frequency is

$$\omega_0 = \frac{2\pi}{p} = \frac{2\pi}{5}.$$

Using the inverse Euler identity,

$$x[n] = \frac{1}{2}e^{i\frac{2\pi}{5}n} + \frac{1}{2}e^{-i\frac{2\pi}{5}n}.$$

Thus, using the coefficient indices $k \in \{-2, -1, 0, 1, 2\}$,

$$X_{-1} = \frac{1}{2}, \quad X_1 = \frac{1}{2},$$

and all other coefficients are zero.

(c) $x[n] = \sin\left(\frac{2\pi}{5}n\right) + \cos\left(\frac{4\pi}{5}n\right), \quad \forall n.$

Answer:

To find the complex-exponential discrete-time Fourier series (DTFS) expansion for $x[n] = \sin\left(\frac{2\pi}{5}n\right) + \cos\left(\frac{4\pi}{5}n\right)$, follow these steps:

1. **Periodicity:** We need to solve for $x[n+p] = x[n]$.

$$\begin{aligned} x[n+p] &= \sin\left(\frac{2\pi}{5}(n+p)\right) + \cos\left(\frac{4\pi}{5}(n+p)\right) \\ &= \sin\left(\frac{2\pi}{5}n + \frac{2\pi}{5}p\right) + \cos\left(\frac{4\pi}{5}n + \frac{4\pi}{5}p\right) \end{aligned}$$

We want this to equal $x[n]$.

$$\begin{aligned} &= \sin\left(\frac{2\pi}{5}n\right) + \cos\left(\frac{4\pi}{5}n\right) \\ &= x[n] \end{aligned}$$

Therefore, we need $\frac{2\pi}{5}p = 2\pi l$ and $\frac{4\pi}{5}p = 2\pi k$, for integers l, k . However, we note that we can write $2\left(\frac{2\pi}{5}p\right) = 2\pi k$, which is the left hand side of the first equation. So, if we solve for the first equation, we'll also solve the second one.

$$\frac{2\pi}{5}p = 2\pi l \iff p = 5l \implies p = 5$$

by setting $l = 1$. Thus, the signal is periodic with the fundamental period $p = 5$. For completeness, plug $p = 5$ into the second equation to get that:

$$\frac{4\pi}{5}(5) = 2\pi k \iff 4\pi = 2\pi k$$

which is satisfied by the integer $k = 2$.

2. **Fundamental Frequency:** The fundamental angular frequency ω_0 is:

$$\omega_0 = \frac{2\pi}{p} = \frac{2\pi}{5}$$

3. **Finding X_k Coefficients using the Analysis Equation:**

$$X_k = \frac{1}{p} \sum_{n=0}^{p-1} x[n] e^{-i\omega_0 kn}$$

where $p = 5$ and $\omega_0 = \frac{2\pi}{5}$.

To avoid going through tedious computations, recognize that we can apply the inverse Euler's identity to cosine and sine:

- The sine term $\sin\left(\frac{2\pi}{5}n\right)$ can be expressed as $\frac{e^{i\frac{2\pi}{5}n} - e^{-i\frac{2\pi}{5}n}}{2i}$.
- The cosine term $\cos\left(\frac{4\pi}{5}n\right)$ can be expressed as $\frac{e^{i\frac{4\pi}{5}n} + e^{-i\frac{4\pi}{5}n}}{2} = \frac{e^{2i\frac{2\pi}{5}n} + e^{-2i\frac{2\pi}{5}n}}{2}$.

The crucial thing to notice here is that these complex exponentials already have $\omega_0 = \frac{2\pi}{5}$ in their exponents. This means that we can read off the coefficients in front of the complex exponentials as the Fourier coefficients. Thus, combining both terms, we can see that the DTFS coefficients X_k are:

$$X_{-2} = \frac{1}{2}, \quad X_{-1} = -\frac{1}{2i}, \quad X_0 = 0, \quad X_1 = \frac{1}{2i}, \quad X_2 = \frac{1}{2}$$

It makes sense that $X_0 = 0$ since there is no constant offset in the signal, ie. no constant term. Don't forget that the analysis equation has a negative in the exponential. Finally, recall that we can choose any range $\langle p \rangle$ for the index k of the Fourier coefficients. In this case, we chose $\{-2, -1, 0, 1, 2\}$

Thus, the DTFS expansion can be written as:

$$x[n] = \frac{1}{2i}e^{i\frac{2\pi}{5}n} - \frac{1}{2i}e^{-i\frac{2\pi}{5}n} + \frac{1}{2}e^{i\frac{4\pi}{5}n} + \frac{1}{2}e^{-i\frac{4\pi}{5}n}$$

In conclusion, we have:

- Fundamental Period $p = 5$
- Fundamental Frequency $\omega_0 = \frac{2\pi}{5}$
- DTFS Expansion:

$$x[n] = \sum_{k=-2}^2 X_k e^{i\frac{2\pi}{5}kn}$$

where X_k are:

$$X_{-2} = \frac{1}{2}, \quad X_{-1} = -\frac{1}{2i}, \quad X_0 = 0, \quad X_1 = \frac{1}{2i}, \quad X_2 = \frac{1}{2}$$

2. Orthogonality of Discrete-Time Complex Exponentials (Fourier Basis)

Let

$$x_k[n] = e^{i\frac{2\pi}{p}kn}, \quad n = 0, 1, \dots, p-1$$

for $k = 0, 1, \dots, p-1$. The complex inner product on $\mathbb{C}^p$ is

$$\langle a, b \rangle = \sum_{n=0}^{p-1} a[n] b^*[n].$$

(a) Show that

$$\langle x_k, x_m \rangle = 0 \quad \text{whenever } k \neq m.$$

Answer: Orthogonality over one period.

$$\langle x_k, x_m \rangle = \sum_{n=0}^{p-1} e^{i\frac{2\pi}{p}kn} e^{-i\frac{2\pi}{p}mn} = \sum_{n=0}^{p-1} e^{i\frac{2\pi}{p}(k-m)n}.$$

Let $k \neq m$, so $k-m \neq 0$. Then this is a finite geometric series:

$$\sum_{n=0}^{p-1} e^{i\frac{2\pi}{p}(k-m)n} = \frac{1 - e^{i\frac{2\pi}{p}(k-m)p}}{1 - e^{i\frac{2\pi}{p}(k-m)}}.$$

Since $e^{i\frac{2\pi}{p}(k-m)p} = e^{i2\pi(k-m)} = 1$,

$$\begin{aligned} \implies \frac{1 - e^{i\frac{2\pi}{p}(k-m)p}}{1 - e^{i\frac{2\pi}{p}(k-m)}} &= \frac{0}{1 - e^{i\frac{2\pi}{p}(k-m)}} = 0 \\ \implies \langle x_k, x_m \rangle &= 0 \quad (k \neq m). \end{aligned}$$

(b) Explain why this orthogonality makes the set

$$\{x_k[n] = e^{i\frac{2\pi}{p}kn} \mid k \in [0, p-1]\}$$

a natural *basis* (the discrete Fourier basis) for $\mathbb{C}^p$.

Answer:

- The family $\{e^{i\frac{2\pi}{p}kn}\}_{k=0}^{p-1}$ is mutually orthogonal by part (a). Each vector has norm $\sqrt{p}$, so dividing each by $\sqrt{p}$ normalizes the vectors without changing their orthogonality.
- The family is mutually orthogonal, so its nonzero vectors are linearly independent. There are exactly p such vectors in the p -dimensional space $\mathbb{C}^p$; therefore, they span $\mathbb{C}^p$ and form a basis.
- This family is called the *discrete Fourier basis* and supplies the basis signals used in DTFS.